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Machine learning and metaheuristic optimization for chiller energy prediction in commercial buildings research trends and future directions

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ABSTRACT

Chiller systems account for a large share of energy use in commercial building HVAC, and accurate prediction supports energy management, demand response, and decarbonization. Although machine learning-based chiller energy prediction has received increasing attention, existing studies have primarily focused on individual prediction models, while the intellectual structure and evolution of research combining machine learning, metaheuristic optimization, and feature selection remain insufficiently mapped. This study conducts a bibliometric analysis of 419 journal articles indexed in the Web of Science Core Collection from 2016 to 2026 using VOSviewer version 1.6.20. The dataset records 20,097 citations, an average of 47.96 citations per paper, and an h-index of 63, with annual output rising from 9 papers in 2016–75 in 2025. Bibliographic coupling of 60 highly cited papers identifies four clusters: ANN-based model predictive control for HVAC, deep and transfer learning for load forecasting, benchmarking of prediction models, and hybrid metaheuristic ANN approaches. Keyword co-occurrence across 56 terms reveals themes in HVAC optimization, machine learning-based load prediction, building energy modeling, and simulation. China leads with 166 publications, representing 39.6 % of total output. The findings establish a comprehensive research map of machine learning and metaheuristic optimization for chiller energy prediction and identify future priorities including chiller-specific feature selection, model interpretability, real-time deployment, climatic generalization, and integration of metaheuristic optimization with deep learning architectures.

1. Introduction

Commercial buildings account for a substantial share of global energy consumption, with heating, ventilation, and air conditioning (HVAC) systems responsible for between 40% and 60% of total building energy use in many countries [1-3]. Among HVAC components, chillers are typically the largest single consumer of electrical energy in commercial buildings, making accurate prediction of chiller power consumption a critical objective for building energy management systems that aim to reduce operational costs and carbon emissions [4-7]. The ability to forecast chiller energy consumption reliably enables building operators to optimize scheduling decisions, implement demand response programs, and evaluate the performance of energy conservation measures in real time [8-10].

The application of machine learning methods to building energy prediction has grown considerably over the past decade, driven by the increasing availability of building operational data from sensor networks and building management systems, and the widespread adoption of accessible computational frameworks for model development [11-13]. Recent systematic reviews have highlighted the increasing role of machine learning in practical building energy management applications, including energy consumption prediction, fault detection, and intelligent control strategies. In parallel, model predictive control has emerged as an important framework for integrating prediction models with HVAC operation optimization, enabling adaptive energy management under changing building conditions [14,15]. Artificial neural networks (ANN) in particular have been extensively applied to chiller energy prediction owing to their ability to model complex nonlinear

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relationships between operational parameters and energy output without requiring explicit physical equations [16,17]. Beyond standard ANN architectures, researchers have explored a broad range of approaches including deep learning, support vector regression, ensemble methods, and hybrid models that combine neural networks with metaheuristic optimization algorithms for feature selection and hyperparameter tuning [18-22].

Feature selection is a particularly important preprocessing step in chiller energy prediction because building management systems collect large numbers of operational and environmental variables, many of which may be redundant, collinear, or only weakly predictive of energy consumption [23-25]. Selecting the most informative input features reduces model complexity, improves generalization to unseen operating conditions, and lowers the computational cost of real-time deployment [26,27]. Metaheuristic optimization algorithms including genetic algorithms, particle swarm optimization, and grey wolf optimizer have been applied to automate this feature selection process within data-driven modeling frameworks [28,29]. Recent studies have also proposed hybrid nature-inspired optimization strategies to improve search capability and model performance in complex engineering applications [30]. The continuous development of nature-inspired optimization algorithms has resulted in new search strategies for solving engineering optimization problems, including parameter tuning and model optimization tasks [31,32].

Despite the rapid growth of individual studies in this domain, the field lacks a systematic and comprehensive mapping of its intellectual structure, thematic evolution, and research frontiers. Conventional narrative literature reviews are inherently limited by subjective paper selection and the inability to quantitatively characterize large bodies of scientific knowledge [33]. Bibliometric analysis provides a rigorous, data-driven, and reproducible alternative, offering the capacity to quantitatively assess publication trends, citation impact, intellectual research fronts, and thematic clusters across large document corpora in a transparent and replicable manner [34].

While recent bibliometric studies have examined artificial intelligence applications in building energy optimization broadly, these reviews [35] generally address the wider building energy domain without isolating chiller systems or examining the specific methodological convergence between machine learning prediction, metaheuristic optimization, and feature selection. Therefore, the intellectual structure, dominant research clusters, and emerging directions of machine learning-based chiller energy prediction remain unclear. The present study directly addresses this gap by providing a comprehensive bibliometric analysis of 419 peer-reviewed journal articles retrieved from the Web of Science Core Collection, covering the period 2016–2026. The analysis employs two complementary VOSviewer-based techniques: bibliographic coupling to map the structural relationships among high-citation documents, and keyword co-occurrence analysis to identify dominant and emerging thematic clusters.

The scientific aim of this study is to establish a comprehensive bibliometric understanding of machine learning and metaheuristic optimization research for chiller energy prediction. To achieve this aim, this study makes the following contributions to the existing literature:

(I) It provides, to the best of our knowledge, the first bibliometric analysis dedicated to the intersection of machine learning, metaheuristic optimization, feature selection, and chiller energy prediction.

(II) It reveals the intellectual structure of the field through bibliographic coupling analysis, identifying four major research clusters that explain the methodological evolution of this research domain.

(III) It maps the thematic landscape through keyword co-occurrence analysis, highlighting dominant research themes, emerging topics, and their relationships.

(IV) It identifies key research gaps and proposes future research directions related to feature selection, model interpretability, real-time deployment, climatic generalization, and hybrid optimization-based learning frameworks.

Unlike previous bibliometric studies that broadly examine artificial intelligence in building energy systems or HVAC applications, this study specifically maps the convergence of machine learning, metaheuristic optimization, feature selection, and chiller energy prediction, thereby providing a focused intellectual framework for future research.

The following research objectives guide this study: (1) to characterize the annual publication and citation growth trends in machine learning-based chiller and building energy prediction research; (2) to identify the geographical distribution of research output and the dominant contributing countries; (3) to delineate the intellectual research fronts through bibliographic coupling network analysis; (4) to identify dominant and emerging thematic clusters through keyword co-occurrence analysis; and (5) to identify critical research gaps and propose future research directions, with particular emphasis on chiller-specific feature selection, model interpretability, and real-world deployment challenges. The remainder of this paper is organized as follows. Section 2 presents the bibliometric methodology and data collection strategy. Section 3 reports the findings of the publication trend, geographical distribution, bibliographic coupling, and co-word analyses. Section 4 discusses the intellectual structure, thematic evolution, research gaps, and implications. Section 5 concludes the study.

2. Methodology

Section 2 describes the bibliometric methodology adopted in this study, including the data collection strategy, search design, and analytical techniques used to identify publication trends, intellectual structures, and thematic patterns in machine learning-based chiller energy prediction research.

2.1. Bibliometric Approach

Bibliometric analysis is a quantitative, data-driven method for evaluating scientific literature through the systematic examination of bibliographic databases using science mapping techniques [34]. This approach enables objective and reproducible characterization of publication trends, citation networks, intellectual structures, and thematic evolution within a research domain, offering a rigorous alternative to conventional narrative literature reviews which are inherently limited by subjective paper selection and the inability to quantitatively characterize large bodies of scientific knowledge [33]. The present study employs VOSviewer (version 1.6.20) as the primary analytical and visualization tool, as this software is particularly robust for handling large document datasets and generating high-density bibliometric network maps [36].

Two complementary bibliometric techniques are employed. First, bibliographic coupling analysis identifies clusters of documents sharing common references, reflecting intellectual similarity and thematic relatedness among publications [37]. Second, keyword co-occurrence analysis (co-word analysis) examines the frequency and joint appearance of author-assigned keywords across the corpus to identify dominant research themes and conceptual structures [38]. Together, these techniques provide a comprehensive understanding of both the structural connections among documents and the thematic evolution of the research domain, enabling holistic mapping of the intellectual landscape of machine learning, metaheuristic optimization, and chiller energy prediction research.

2.2. Data Collection and Search Strategy

The bibliographic data were retrieved from the Web of Science (WoS) Core Collection, widely recognized as one of the most authoritative databases for bibliometric research [39]. The search was conducted on 11 March 2026 using an advanced topic search applied across Title, Abstract, and Keywords via the Topic (TS) field. All searches were performed within the Web of Science Core Collection using the Topic

(TS) field, which simultaneously searches article titles, abstracts, author keywords, and Keywords Plus. No manual screening or supplementary database searches were performed. The search strategy was developed iteratively through a series of refinement trials to balance specificity and coverage, arriving at two complementary search strings subsequently merged using a Boolean OR operator.

Search String A was constructed around four conceptual pillars: (1) building or commercial building contexts; (2) machine learning and neural network methods; (3) chiller or cooling load systems; and (4) prediction or forecasting tasks:

("building energy" OR "commercial building") AND ("machine learning" OR "neural network" OR "artificial neural network") AND ("chiller" OR "cooling load" OR "chiller system") AND ("prediction" OR "forecasting")

.

Search String C additionally captured studies at the intersection of HVAC systems, machine learning, energy prediction, and optimization or feature selection:

("chiller" OR "chiller system" OR "HVAC") AND ("neural network" OR "machine learning") AND ("energy prediction" OR "power prediction" OR "energy consumption") AND ("optimization" OR "feature selection")

String A independently retrieved 148 documents, while String C retrieved 288 documents. The Boolean OR merge of both strings yielded a combined dataset of 419 documents spanning 2016–2026. After restricting the document type to peer-reviewed journal articles published in English only, the corpus remained at 419 documents. The search results were restricted to peer-reviewed journal articles published in English. Review papers, conference proceedings, book chapters, editorial materials, meeting abstracts, and other document types were excluded to ensure a consistent dataset for bibliometric analysis. Table 1 summarizes the key parameters and statistics of the final dataset.

The selection of keywords was guided by the specific research scope of chiller energy prediction in commercial buildings. Terms related to the building context, such as "building energy" and "commercial building", were included to capture studies associated with building energy management. The terms "chiller", "cooling load", and "chiller system" were selected to distinguish chiller-focused prediction studies from broader HVAC research. Machine learning-related terms, including "machine learning", "neural network", and "artificial neural network", were included to represent data-driven prediction approaches, while prediction-related terms were incorporated to identify forecasting and estimation studies.

Broader thermal system terms, such as "heat pump" and "refrigeration", were not included because they represent wider application areas with different operating characteristics and research objectives. Including these terms could increase the number of retrieved publications but introduce studies outside the focus of chiller energy prediction. Similarly, optimization-related terms were included only in String C to capture studies involving feature selection and parameter optimization associated with machine learning-based prediction models. This keyword design aimed to achieve a balance between retrieval coverage and thematic relevance.

Table 1
Search string parameters and dataset statistics.

Parameter	Details	Value
Database	Web of Science Core Collection	WoS CC
Search date	11 March 2026	—
Search fields	Title, Abstract, Keywords (Topic field)	TS=
String A	("building energy" OR "commercial building") AND ("machine learning" OR "neural network" OR "artificial neural network") AND ("chiller" OR "cooling load" OR "chiller system") AND ("prediction" OR "forecasting")	148 docs
String C	("chiller" OR "chiller system" OR "HVAC") AND ("neural network" OR "machine learning") AND ("energy prediction" OR "power prediction" OR "energy consumption") AND ("optimization" OR "feature selection")	288 docs
Merged (A OR C)	Boolean OR merge of String A and String C	419 docs
Document type	Article	—
Language	English	—
Final corpus	Peer-reviewed journal articles, 2016–2026	419 docs
Year range	2016–2026	11 years
Total citations	Times Cited (all databases)	20,097
H-index	Based on WoS citation report	63
Avg. citations/doc	Total citations divided by number of publications	47.96
Analysis tool	VOSviewer version 1.6.20	—
Techniques	Bibliographic coupling; Keyword co-occurrence (co-word analysis)	Two methods

Source: Web of Science Core Collection, 11 March 2026

2.3. Research Domain and Subject Categories

The 419 documents in the final corpus span multiple Web of Science subject categories, reflecting the inherently multidisciplinary nature of chiller energy prediction research. As illustrated in Fig. 1, the dominant subject category is Energy Fuels (n = 216), confirming that chiller energy prediction is primarily framed within the energy engineering literature, followed by Construction Building Technology (n = 169), Engineering Civil (n = 151), Thermodynamics (n = 68), Engineering Electrical Electronic (n = 40), Green Sustainable Science Technology (n = 37), Engineering Chemical (n = 33), Computer Science Information Systems (n = 25), Engineering Mechanical (n = 24), and Engineering Environmental (n = 18), among others. This cross-disciplinary distribution confirms that chiller power prediction occupies the intersection of energy engineering, building science, thermodynamics, and computational intelligence, a profile consistent with the scope of *Energy* (Elsevier).

Further classification using the WoS Citation Topics hierarchy provides a more granular view of the research domain. At the Meso level, the corpus is overwhelmingly classified under 6.115 Sustainability Science (n = 369), confirming the strong sustainability and energy efficiency orientation of the field. Secondary Meso classifications include 4.18 Power Systems & Electric Vehicles (n = 12), 7.70 Thermodynamics (n = 8), 4.46 Distributed & Real Time Computing (n = 3), and 4.224 Design & Manufacturing (n = 2). At the Micro level, the dominant topic is 6.115.284 Building Energy Efficiency (n = 368), which directly confirms that the overwhelming majority of documents in this corpus are specifically aligned with building energy efficiency research — precisely the application domain of the present bibliometric study. Additional Micro topics include 4.18.296 Energy Forecasting (n = 4) and 7.70.1455 Liquid Desiccant (n = 4), further reinforcing the energy systems and HVAC orientation of the dataset. Table 2 presents the complete Citation Topics Meso and Micro classification for the dataset.

In terms of publisher distribution, Elsevier dominates the corpus with 257 publications (approximately 61.3% of all documents), followed by MDPI (n = 87), IEEE (n = 21), Springer Nature (n = 14), and Taylor & Francis (n = 9). The strong Elsevier representation reflects the concentration of high-impact building energy and HVAC journals within its portfolio, including *Energy and Buildings*, *Applied Energy*, *Energy*, and the *Journal of Building Engineering*, which collectively serve as the primary publication venues for chiller energy prediction research.

Regarding open access status, of the 419 documents, 193 articles (46.1%) are available under some form of open access, of which 126 (30.1%) are published as Gold Open Access, 39 (9.3%) as Gold-Hybrid, 8 (1.9%) as Green Published, 6 (1.4%) as Free to Read, and 5 (1.2%) as Green Accepted. An additional 139 documents (33.2%) are available as Green Submitted preprints. The remaining 226 documents (53.9%) are

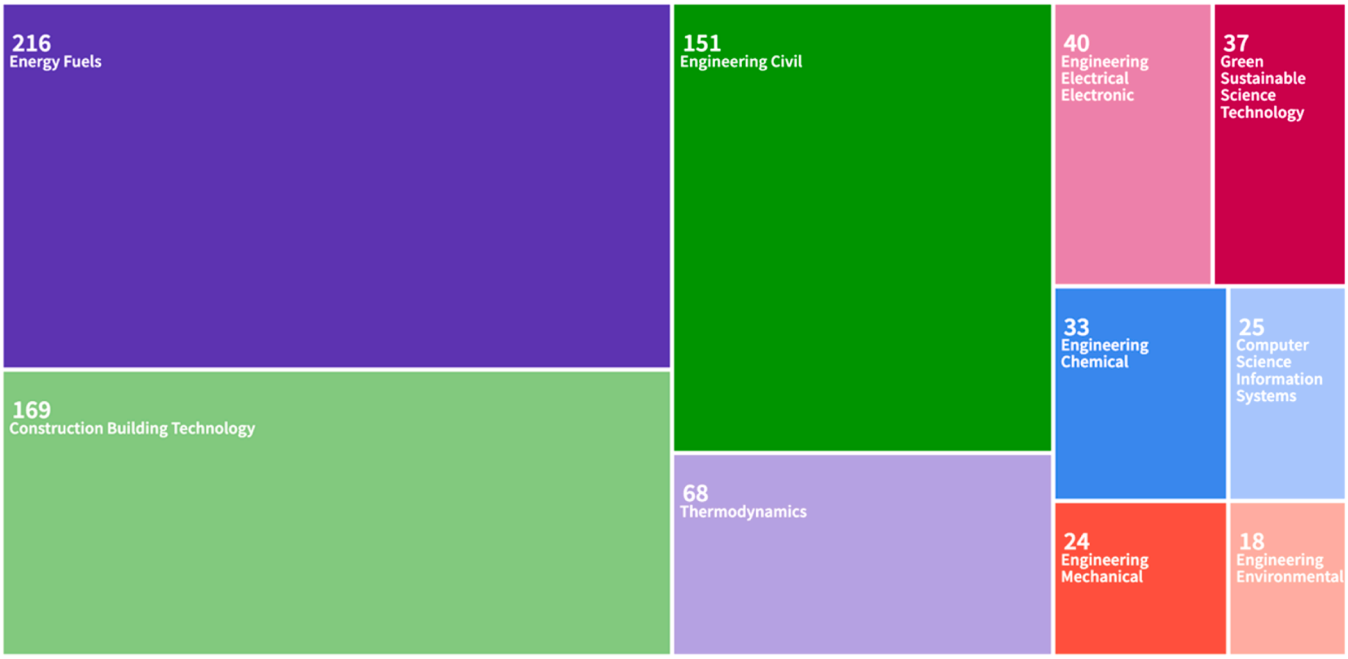


Fig. 1. TreeMap chart for the publications field of study from 2016 to 2026. (Source: Web of Science)

Table 2
WoS Citation Topics Meso and Micro classification of the dataset.

Level	Code	Topic	No. of Documents
Meso	6.115	Sustainability Science	369
Meso	4.18	Power Systems & Electric Vehicles	12
Meso	7.70	Thermodynamics	8
Meso	4.46	Distributed & Real Time Computing	3
Meso	4.224	Design & Manufacturing	2
Micro	6.115.284	Building Energy Efficiency	368
Micro	4.18.788	Electric Vehicles	5
Micro	4.18.296	Energy Forecasting	4
Micro	7.70.1455	Liquid Desiccant	4
Micro	4.46.85	Cloud Resource Scheduling	3

Source: Web of Science Core Collection, March 2026

accessible only through subscription-based channels. The substantial open access proportion reflects the growing adoption of open science mandates by funding agencies and institutions globally, particularly in the energy and engineering research communities. Table 3 summarizes the full open access distribution of the dataset.

2.4. Publication Trend and Citation Analysis

Annual publication and citation trends were examined to assess the growth trajectory of the field over the 2016–2026 period. As depicted in Fig. 2, the publication trend reveals a clear and sustained upward trajectory, growing from 9 articles in 2016 to a peak of 75 articles in 2025, with a notable acceleration from 2019 onwards. Intermediate growth milestones include 17 articles in 2018, 29 in 2019, 34 in 2020, 40 in 2021, 55 in 2022, 65 in 2023, and 63 in 2024, before peaking at 75 in 2025. The year 2026 shows only 18 articles at the time of data collection, reflecting the partial nature of the ongoing publication year rather than a decline in research activity. This sustained growth pattern is consistent with broader global trends in building energy research, driven by increasing urgency around commercial building energy efficiency, widespread adoption of machine learning frameworks, and intensifying policy pressures related to decarbonization and net-zero targets for the built environment.

Table 3
Open access distribution of the 419-document corpus.

Open Access Category	Number of Documents	Percentage (%)
All Open Access	193	46.1
— Gold Open Access	126	30.1
— Gold-Hybrid	39	9.3
— Free to Read	6	1.4
— Green Published	8	1.9
— Green Accepted	5	1.2
Green Submitted	139	33.2
Non-Open Access (Subscription only)	226	53.9
Total	419	100.0

Source: Web of Science Core Collection, March 2026

The citation trajectory closely mirrors the publication growth, with cumulative citations reaching 20,097 total times cited across all databases, and 18,816 citations excluding self-citations. The average citation rate stands at 47.96 citations per document, and the dataset achieved an h-index of 63, indicating a substantial and highly influential body of work. The sharp citation growth from 2020 onwards reflects the increasing adoption of deep learning and hybrid metaheuristic approaches in building energy research, where foundational works proposing ANN-based chiller models and feature selection frameworks have become widely cited reference points. The apparent decline in the citation curve for 2026 is an artifact of incomplete annual data at the time of retrieval, consistent with standard bibliometric reporting conventions.

The geographical distribution of publications reveals a pronounced concentration of output across 66 countries. China (People's Republic) dominates with 166 publications (39.6% of the corpus), reflecting its position as the world's largest commercial building market and its substantial investment in building energy management and artificial intelligence research. The United States follows with 51 publications, South Korea with 30, Iran with 29, Spain with 24, Canada and England with 21 each, and Saudi Arabia with 20. India contributes 19 publications, Singapore 15, Italy 13, and Australia 12. Malaysia contributes 8 publications, demonstrating the growing engagement of Southeast Asian institutions in this domain. The broad representation across 66 countries confirms the global relevance and interdisciplinary reach of machine

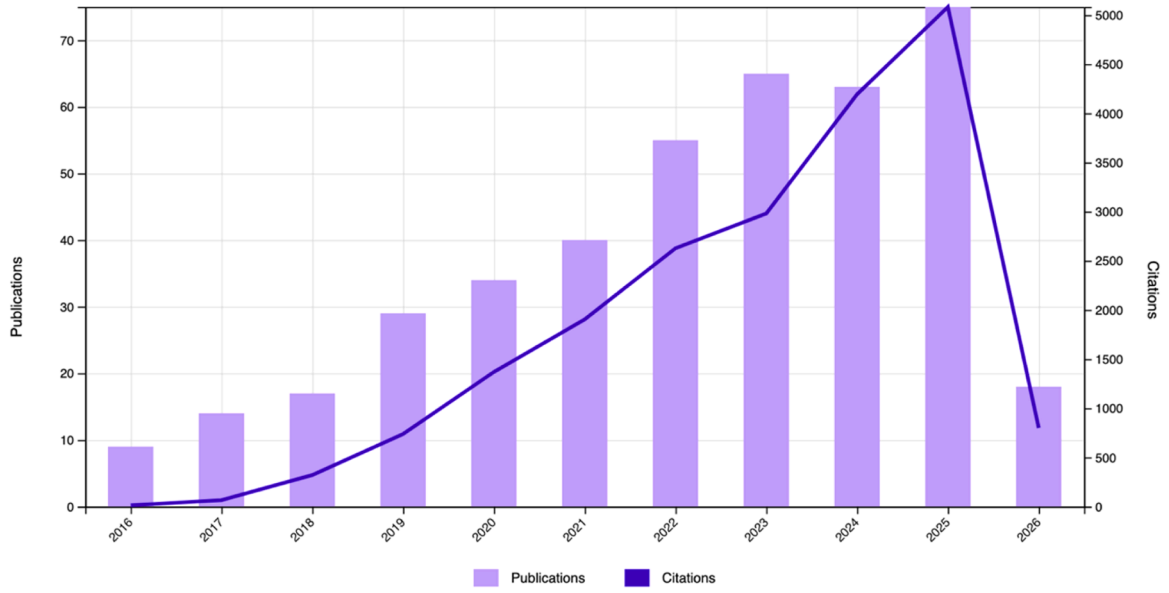


Fig. 2. Number of publications and citations from 2016 to 2026.
(Source: Web of Science)

learning and optimization research applied to chiller energy systems in commercial buildings.

3. Findings

structure and thematic development of machine learning and metaheuristic optimization research for chiller and building energy prediction. Bibliographic coupling identifies groups of publications with shared references, revealing common research foundations and topics, while co-word analysis examines keyword co-occurrence to identify major themes and emerging trends. Both analyses were performed using VOSviewer (version 1.6.20), enabling the visualization of research networks and providing a comprehensive understanding of the field's structural and thematic evolution.

3.1. Bibliographic Coupling

The following threshold and clustering parameters were applied in VOSviewer for the bibliographic coupling analysis. A minimum citation threshold of 68 was applied, of which 60 of the 419 documents met this criterion. All 60 items were connected within the network with no unconnected items excluded. Clustering was performed at the default resolution of 1.00 with a minimum cluster size of 1 and the merge small clusters option enabled. The resulting network comprised 60 items distributed across 4 distinct clusters, connected by 976 links with a total link strength of 3772. The node size in the VOSviewer map represents the number of citations of each document, while the thickness of connecting lines reflects the strength of bibliographic coupling between document pairs. The four clusters represent distinct but interrelated research fronts within the domain of machine learning and optimization applied to building energy and chiller systems.

3.1.1. Network Visualization of Bibliographic Coupling

The bibliographic coupling network generated by VOSviewer is exhibited in Fig. 3. The network reveals a well-connected and densely interconnected body of literature with a prominent core of highly influential documents. [40,41], and [42] dominate the central region of the network by node size and connectivity, indicating their roles as the most widely referenced intellectual anchors across the entire corpus. The four distinct clusters are visually well-separated while maintaining

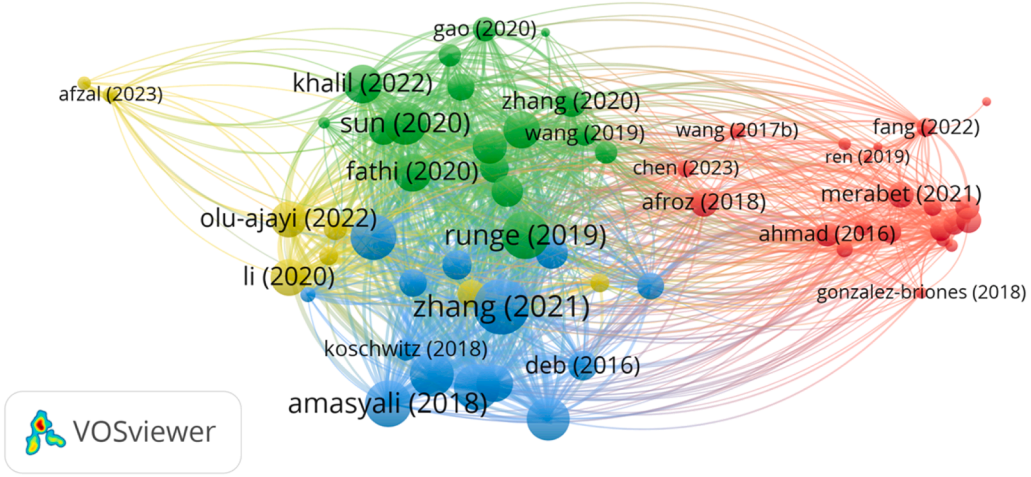


Fig. 3. Network visualization of bibliographic coupling of documents within the research field.

inter-cluster connections, confirming the existence of distinct research fronts that share common methodological foundations.

Table 4 presents the top 10 documents ranked by total link strength (TLS) in the bibliographic coupling network. [40] emerged as the most strongly coupled document with a TLS of 472 and 449 citations, followed by [42] with a TLS of 348 and [41] with a TLS of 335. Notably, [41] recorded the highest citation count in the entire dataset at 1340, reflecting its standing as the most comprehensive and widely referenced review of data-driven building energy consumption prediction in the field. [43] and [44] follow with TLS values of 324 and 321 respectively, while [45] and [46] share an equal TLS of 286. [47] and [48] represent more recent contributions that have achieved strong bibliographic coupling despite their relatively short citation accumulation periods, indicating rapid uptake into the intellectual core of the field.

3.1.2. Cluster 1 (Red): ANN-Based Model Predictive Control and Optimization of HVAC Systems

Cluster 1 (red) is the largest cluster, comprising 21 documents, and focuses on the use of artificial neural network (ANN) based model predictive control (MPC) and optimization frameworks for HVAC systems. The core theme across this cluster is the use of ANN models as active components within closed-loop control architectures that optimize HVAC setpoints to minimize energy consumption while maintaining indoor environmental quality. [49] provided one of the earliest comprehensive reviews of ANN based MPC for HVAC applications, introducing the Best Network after Multiple Iterations algorithm for optimal ANN architecture selection and demonstrating operating cost reductions of between 6% and 73% relative to fixed setpoint control. [50] extended this by integrating a NARX neural network with particle swarm optimization to simultaneously minimize HVAC energy use and maintain indoor temperature, humidity, CO2, and VOC levels within ASHRAE Standard 55 limits, achieving a 7.8% energy reduction without compromising occupant comfort. [51] applied ANN based MPC to a sports facility at Qatar University, achieving up to 46% energy reduction while optimizing thermal comfort and indoor air quality, with the ANN model outperforming SVR, kNN, and decision tree alternatives. Other documents in this cluster, including [52-55], and [56], further address ANN based control optimization across diverse building types and climatic conditions. Collectively, this cluster establishes ANN based intelligent control as a validated approach for HVAC energy optimization, providing the operational context for the predictive modeling focus of the present bibliometric study.

3.1.3. Cluster 2 (Green): Deep Learning and Transfer Learning for Building Energy Load Forecasting

Cluster 2 (green) consists of 17 documents and focuses on deep learning and transfer learning architectures for short-term and medium-term building energy load forecasting. [57] conducted a systematic assessment of deep recurrent neural network-based strategies for short-term building energy prediction, evaluating recursive, direct, and multi-input multi-output inference approaches combined with convolutional operations and bidirectional architectures, providing a benchmark for practitioners developing deep recurrent models from building operational data. [58] addressed energy prediction under limited data conditions by proposing a sequence-to-sequence model and a convolutional neural network with attention layer, combined with a transfer learning framework for newly constructed or data-scarce buildings, achieving improvements of approximately 19.69 and 20.54 %age points in MAPE relative to a standard LSTM baseline. [59] proposed a hybrid LSTM-ANN method for short-term cooling load prediction that incorporated a dimensionless sensitivity index and weighted Manhattan distance to improve both prediction accuracy and the interpretability of key operational drivers. [42,46,47], and [48], all among the top 10 most strongly coupled documents in the network, contribute additional work on comparative evaluation of ML architectures, real-world data integration, and uncertainty-aware deep learning for building energy systems. Collectively, this cluster reflects the maturation of deep learning for building energy load forecasting, and its strong coupling with Clusters 1 and 3 highlights the shared methodological foundations connecting intelligent control, prediction, and optimization research in this domain.

Table 4
Top 10 documents in bibliographic coupling analysis.

Rank	Document	Citations	TLS	Cluster
1	Zhang (2021)	449	472	Cluster 3 (Blue)
2	Runge (2019)	197	348	Cluster 2 (Green)
3	Amasyali (2018)	1340	335	Cluster 3 (Blue)
4	Wang (2020)	201	324	Cluster 3 (Blue)
5	Ahmad (2018)	270	321	Cluster 3 (Blue)
6	Wang (2017a)	498	301	Cluster 3 (Blue)
7	Deb (2017)	714	286	Cluster 3 (Blue)
8	Sun (2020)	385	286	Cluster 2 (Green)
9	Khalil (2022)	202	232	Cluster 2 (Green)
10	Fathi (2020)	220	226	Cluster 2 (Green)

Note: TLS = Total Link Strength. Source: VOSviewer analysis of WoS dataset, 11 March 2026.

3.1.4. Cluster 3 (Blue): Review and Benchmarking of ML and Data-Driven Building Energy Prediction Methods

Cluster 3 (blue) contains 14 documents and represents the most intellectually foundational cluster in the network, anchored by comprehensive review and benchmarking studies on machine learning and data-driven methods for building energy prediction. [40], the most strongly coupled document in the entire network (TLS = 472), provided a landmark review of machine learning applications in building load prediction organized around the task-performance-experience framework, systematically examining prediction tasks, modeling algorithms, data engineering, feature selection methods, and future research directions. [41], with the highest citation count in the dataset (1340 citations), offered a complementary review focused on data-driven building energy consumption prediction, analyzing prediction scopes, preprocessing methods, machine learning algorithms, and performance metrics across a large body of prior studies, and identifying research gaps that have since shaped the field's agenda. [60] contributed an empirical benchmarking study comparing support vector machine regression against NARX recurrent neural networks for heating and cooling load prediction at district scale in Germany, finding that NARX models achieved higher accuracy with comparable computational effort. [43-45,61] and [62] further contribute foundational work on ANN training strategies, gradient-based and evolutionary optimization, and support vector machine applications in HVAC load prediction. The high citation counts across this cluster, including [41] at 1340, [45] at 714, [61] at 498, and [40] at 449, confirm that Cluster 3 represents the shared intellectual foundation of data-driven building energy prediction research and directly supports the motivation of the present study.

3.1.5. Cluster 4 (Yellow): Hybrid ML-Metaheuristic Optimization for Heating and Cooling Load Prediction

Cluster 4 (yellow) comprises 8 documents and is the cluster most directly aligned with the methodological contribution of the present study, grouping hybrid approaches that combine machine learning models, particularly multilayer perceptron (MLP) and ANN architectures, with metaheuristic optimization algorithms for building heating and cooling load prediction. [63] proposed an MLP framework hybridized with eight metaheuristic algorithms including particle swarm optimization, grey wolf optimizer, and a hybrid PSO-GWO variant, with the hybrid MLP-PSOGWO model achieving R^2 values of 0.966 for cooling load and 0.998 for heating load, demonstrating that metaheuristic hyperparameter optimization substantially improves MLP prediction performance. [64] extended this work with additional comparative analysis of optimization strategies for neural network-based building

energy prediction. [65] developed a machine learning approach for predicting annual residential heating and cooling loads that incorporated occupant behavior as a predictor variable, evaluating five models including support vector regression and ANN, with Gaussian RBF kernel SVR achieving less than 4% normalized MAE and RMSE. [66] proposed extreme learning machine and online sequential extreme learning machine models for heating and cooling load prediction from building design attributes, developing 24 models with different feature sets and activation functions and demonstrating that extreme learning machine approaches outperform ANN, SVM, and random forest baselines in both accuracy and computational efficiency. [67] and [68] contribute further perspectives on ensemble ML methods and real-time adaptive prediction for building performance assessment.

Table 5 presents a summary of the bibliographic coupling analysis including cluster number and color, thematic label, number of publications, and representative documents.

3.2. Co-word Analysis (Keyword Co-occurrence)

Keyword co-occurrence analysis, also referred to as co-word analysis, was performed to identify the dominant research themes and conceptual structure within the 419-document corpus. In this type of analysis, two keywords are considered co-occurring when they appear together within the same document, and the strength of their co-occurrence reflects the frequency of their joint appearance across the dataset [38]. Co-word analysis complements bibliographic coupling by revealing the intellectual content and thematic evolution of the field rather than its citation-based structural relationships.

Of the 1734 unique keywords identified across all documents, a minimum occurrence threshold of 16 was applied, resulting in 56 keywords meeting the threshold for inclusion in the network. The co-word network was constructed using VOSviewer (version 1.6.20) with the association strength normalization method. Clustering parameters were set at a resolution of 1.05 and a minimum cluster size of 1, with the merge small clusters option enabled. The resulting network comprised 56 items distributed across 4 thematic clusters, connected by 1257 links with a total link strength of 5106. Table 6 summarizes the top 10 keywords ranked by total link strength.

Table 5
Bibliographic coupling analysis — cluster summary.

Cluster	Colour	Items	Theme Label	Representative Documents
1	Red	21	ANN-Based Model Predictive Control and Optimization of HVAC Systems	Afram (2017), Afroz (2022), Elnour (2022), Yang (2020), Yao (2021)
2	Green	17	Deep Learning and Transfer Learning for Building Energy Load Forecasting	Fan (2019), Gao (2020), Zhang (2020), Runge (2019), Sun (2020), Khalil (2022)
3	Blue	14	Review and Benchmarking of ML and Data-Driven Building Energy Prediction Methods	Zhang (2021), Amasyali (2018), Koschwitz (2018), Wang (2020), Deb (2017)
4	Yellow	8	Hybrid ML-Metaheuristic Optimization for Heating and Cooling Load Prediction	Afzal (2023), Afzal (2024), Li (2020), Kumar (2018), Olu-Ajayi (2022)

Note: Of the 419 documents in the dataset, 60 met the minimum citation threshold of 68 citations. All 60 items were successfully mapped into the network (Links: 976; Total link strength: 3772). Clustering was performed using the association strength normalization method with a resolution of 1.00, minimum cluster size of 1, and merge small clusters option enabled.

Table 6
Top 10 keywords in the co-occurrence analysis.

Rank	Keyword	Occurrences	TLS	Cluster
1	optimization	138	646	Cluster 1 (Red)
2	machine learning	104	472	Cluster 2 (Green)
3	performance	97	459	Cluster 4 (Yellow)
4	model	76	416	Cluster 3 (Blue)
5	artificial neural network	68	367	Cluster 2 (Green)
6	cooling load	61	330	Cluster 2 (Green)
7	thermal comfort	64	325	Cluster 1 (Red)
8	energy-consumption	65	297	Cluster 4 (Yellow)
9	simulation	50	284	Cluster 4 (Yellow)
10	consumption	45	265	Cluster 3 (Blue)

Note: TLS = Total Link Strength; Occurrences = number of documents containing the keyword. Source: VOSviewer analysis of WoS dataset, 11 March 2026.

3.2.1. Network Visualization of Keyword Co-occurrence

The co-word network map revealed four distinct thematic clusters reflecting the major conceptual streams within the literature on machine learning, optimization, and chiller and HVAC energy systems, as depicted in Fig. 4. The network is densely connected with a high total link strength of 5106, indicating strong conceptual interdependence across research themes. Central nodes such as 'optimization', 'machine learning', 'model', and 'neural-network' occupy prominent bridging positions in the network, connecting multiple clusters and underscoring their cross-cutting relevance to the entire research domain. The dominance of 'optimization' (occurrences = 138, TLS = 646) as the highest-frequency and most strongly connected term in the entire network confirms that this research corpus is fundamentally oriented toward improving the efficiency and performance of building energy systems through computational optimization methods.

3.2.2. Cluster 1 (Red): HVAC Energy Optimization, Metaheuristic Algorithms, and Intelligent Control

Cluster 1 (red) is the largest thematic cluster, comprising 25 keywords, and represents the dominant research stream on HVAC energy optimization using metaheuristic algorithms and intelligent control strategies. The cluster is anchored by 'optimization' (occurrences = 138, TLS = 646), the highest-ranked keyword in the entire network, reflecting the central role of optimization as both a research objective and a computational tool in this domain. Co-occurring terms include 'genetic algorithm', 'multiobjective optimization', 'energy optimization', 'model predictive control', and 'predictive control', confirming that this cluster spans classical evolutionary optimization through to modern model-based control strategies for HVAC energy management. The co-occurrence of 'thermal comfort' (occurrences = 64, TLS = 325) alongside optimization and control keywords reflects a defining challenge in this field: minimizing HVAC energy consumption while maintaining acceptable indoor thermal conditions, a problem that single-objective optimization methods alone cannot resolve. The presence of 'demand response', 'hvac system', 'ventilation', 'management', 'buildings', and 'efficiency' within the same cluster confirms the applied engineering orientation of this research stream, where optimization is deployed within real building management systems rather than purely theoretical settings. This cluster directly corresponds to Cluster 1 (red) in the bibliographic coupling analysis, where ANN-based MPC and HVAC optimization papers including [49,50], and [51] were grouped together, confirming strong conceptual alignment between the document-level and keyword-level analyses.

3.2.3. Cluster 2 (Green): ANN and ML-Based Cooling and Heating Load Prediction

Cluster 2 (green) contains 20 keywords and represents the core methodological cluster focused on the application of artificial neural networks and machine learning for cooling and heating load prediction

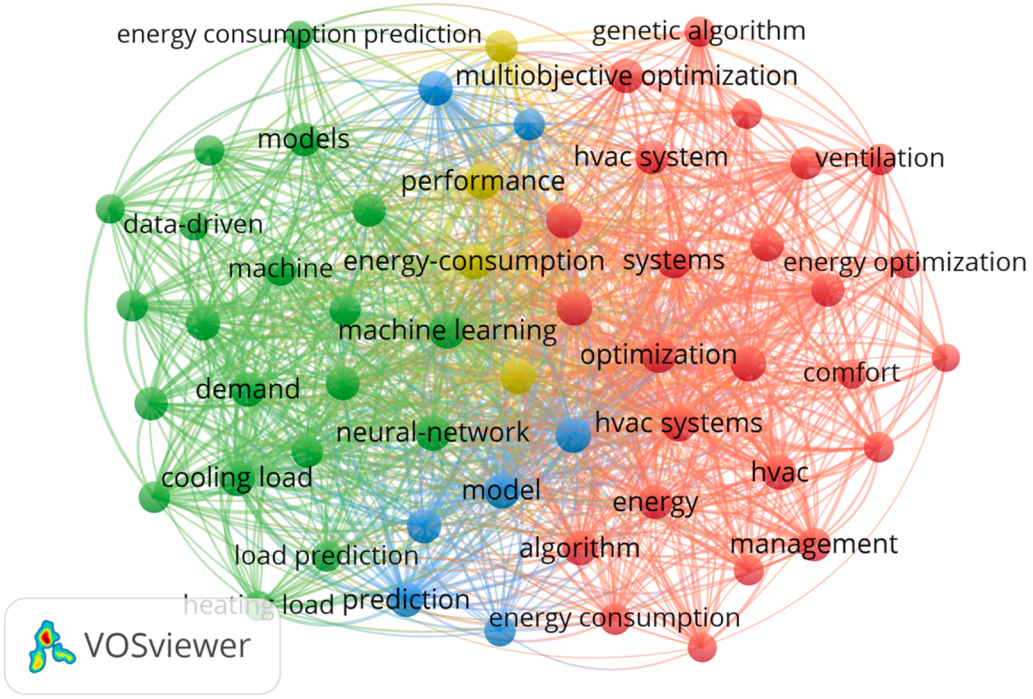


Fig. 4. Bibliometric map of keyword co-occurrences.

in buildings. The cluster is anchored by 'machine learning' (occurrences = 104, TLS = 472) and 'artificial neural network' (occurrences = 68, TLS = 367), both among the top five most strongly connected keywords in the entire network. The co-occurrence of 'cooling load' (occurrences = 61, TLS = 330), 'cooling load prediction', 'heating load', 'load prediction', and 'demand' confirms that thermal load forecasting across both cooling and heating applications is the primary prediction task in this cluster. The presence of 'deep learning', 'data-driven', 'random forest', 'support vector regression', and 'regression' reflects methodological diversity spanning classical machine learning, ensemble methods, and deep learning architectures applied to the same core prediction problem. The co-occurrence of 'office buildings' and 'residential buildings' indicates that load prediction studies in this cluster address multiple building typologies, while 'electricity consumption' and 'energy consumption prediction' confirm the energy-specific framing of prediction tasks.

3.2.4. Cluster 3 (Blue): ANN Model Development for Building Energy Consumption Prediction

Cluster 3 (blue) consists of 7 keywords and represents a compact but conceptually distinct grouping centered on ANN model development for building energy consumption prediction. The cluster comprises 'artificial neural network', 'building energy', 'consumption', 'model', 'prediction', and 'system'. Despite its small size, this cluster occupies a central bridging position in the co-word network, with 'model' (occurrences = 76, TLS = 416) and 'prediction' (occurrences = 49, TLS = 252) serving as high-connectivity nodes that link Clusters 1, 2, and 4. The co-occurrence of 'building energy', 'consumption', and 'system' with ANN terms confirms that this cluster captures the applied modeling literature where ANN architectures are developed and validated specifically for whole-building energy consumption prediction using operational data from real building management systems. The tight conceptual coherence of this small cluster suggests that these terms consistently appear together as a core vocabulary across the corpus, functioning as conceptual anchors that connect the optimization-oriented vocabulary of Cluster 1 with the broader machine learning vocabulary of Cluster 2. The central network position of this cluster confirms that building energy prediction using ANN remains a foundational and unifying theme across the entire research domain, regardless of the specific optimization method or building type addressed in individual studies.

3.2.5. Cluster 4 (Yellow): Building Performance Simulation and Energy Consumption Modelling

Cluster 4 (yellow) is the smallest cluster, comprising 4 keywords: 'design', 'energy consumption', 'performance', and 'simulation'. This cluster represents a distinct research stream focused on building performance simulation and energy consumption modeling from a design and systems engineering perspective. The term 'performance' (occurrences = 97, TLS = 459) is the third most strongly connected keyword in the entire network, indicating that building and system performance assessment is a consistent concern across the full corpus. The co-occurrence of 'simulation', 'design', and 'energy consumption' reflects the simulation-based strand of building energy research, where energy consumption is modeled at the design stage using tools such as EnergyPlus or TRNSYS, rather than predicted from live operational data as in Clusters 2 and 3. The term 'energy consumption' (occurrences = 65, TLS = 297) also functions as a bridging node between this cluster and Clusters 1 and 2, reflecting its dual role as both a simulation output metric and a machine learning prediction target across the broader literature. The presence of this cluster confirms that simulation-based energy modeling remains an active and complementary strand of building energy research, providing baseline benchmarks and physics-informed inputs that increasingly support and inform data-driven prediction approaches.

A summary of the co-word analysis is presented in Table 7, comprising cluster number and color, cluster labels, number of keywords, and representative keywords.

4. Discussion

This section discusses the findings from the bibliometric analyses by interpreting the intellectual structure, thematic evolution, research gaps, and practical implications of machine learning and optimization approaches for chiller energy prediction.

Table 7
Co-word analysis — thematic clusters.

Cluster	Colour	Items	Theme Label	Representative Keywords
1	Red	25	HVAC Energy Optimization, Metaheuristic Algorithms, and Intelligent Control	optimization, genetic algorithm, multiobjective optimization, HVAC system, HVAC systems, thermal comfort, model predictive control, energy optimization, demand response, ventilation, comfort, management
2	Green	20	ANN and ML-Based Cooling and Heating Load Prediction	machine learning, artificial neural-network, cooling load, cooling load prediction, deep learning, heating load, load prediction, data-driven, neural-network, random forest, support vector regression, demand
3	Blue	7	ANN Model Development for Building Energy Consumption Prediction	artificial neural network, building energy, consumption, model, prediction, system
4	Yellow	4	Building Performance Simulation and Energy Consumption Modelling	design, energy-consumption, performance, simulation

Note: Of the 1734 keywords identified, 56 met the minimum occurrence threshold of 16. All 56 items were successfully mapped into the network (Links: 1257; Total link strength: 5106). Clustering was performed using the association strength normalization method with a resolution of 1.05, a minimum cluster size of 1, and the merge small clusters option enabled.

4.1. Publication Trend Analysis

The annual publication trend confirms that research on machine learning and optimization for building energy and chiller systems has grown substantially over the past decade, rising from 9 articles in 2016 to a peak of 75 articles in 2025. This trajectory is consistent with the wider expansion of data-driven methods in building engineering research, which accelerated following the broad adoption of deep learning frameworks and the increasing availability of building operational data from sensor networks and building management systems [41]. The acceleration observed from 2019 onwards coincides with growing policy pressure on the built environment sector to reduce energy consumption, particularly through the tightening of building energy codes and the inclusion of commercial buildings in national net-zero commitments in many countries [69].

The citation metrics further reflect the maturity and influence of this research domain. The dataset achieved an h-index of 63 with a total of 20,097 citations and an average of 47.96 citations per document, which is high relative to many applied engineering subfields. The citation growth observed from 2020 onwards reflects the increasing uptake of foundational review papers, particularly [41] with 1340 citations and [40] with 449 citations, both of which have become standard reference points for methodological justification in building energy prediction studies. The partial publication and citation data for 2026 reflect the ongoing nature of the current year at the time of data retrieval and should not be interpreted as a slowdown in research activity.

The dominance of China in research output, accounting for 166 publications (39.6% of the corpus), is consistent with patterns observed in other building energy bibliometric analyses and reflects China's position as both the world's largest commercial building market and a major investor in artificial intelligence research infrastructure [34]. The United States (51 publications), South Korea (30), Iran (29), and Spain (24) constitute the next tier of contributing countries, while the presence of 66 countries in total indicates that this research domain has a broad global reach rather than being concentrated in a small number of systems. The relatively modest contribution of Southeast Asian countries, including Malaysia with 8 publications, suggests that there is scope for expanded research engagement from institutions in this region, particularly given the high cooling energy demand associated with tropical climates where chiller systems operate year-round.

4.2. Intellectual Structure: Bibliographic Coupling Clusters

The bibliographic coupling analysis identified four interconnected research clusters that illustrate the evolution of the field from foundational machine learning studies to advanced hybrid optimization approaches. The review and benchmarking cluster forms the intellectual foundation of the field, providing common evaluation frameworks and methodological references for subsequent research. Another major cluster focuses on ANN-based model predictive control and HVAC optimization, highlighting the importance of accurate energy prediction for improving operational efficiency and reducing building energy costs.

The four clusters also reveal a chronological and methodological progression within the research domain. The field initially developed around establishing reliable prediction frameworks and benchmarking practices, represented by the review and benchmarking studies in Cluster 3. As operational data availability increased through building sensors and management systems, research expanded toward ANN-based control and optimization strategies in Cluster 1. The subsequent adoption of advanced computational architectures led to the emergence of deep learning and transfer learning approaches in Cluster 2, which aimed to capture complex temporal patterns in energy consumption data. More recently, Cluster 4 reflects a shift toward hybrid optimization frameworks, where metaheuristic algorithms are integrated with machine learning models to improve feature selection, parameter tuning, and prediction performance.

The remaining clusters represent methodological advances in building energy forecasting. The emergence of deep learning and transfer learning as a distinct cluster reflects the shift from conventional shallow learning models toward architectures capable of capturing temporal dependencies in energy consumption data. Recent studies have demonstrated the effectiveness of LSTM-based and hybrid deep learning approaches for forecasting complex energy profiles, although their application to chiller-specific prediction remains constrained by data availability, computational requirements, and limited interpretability. This finding suggests that future research should focus not only on improving prediction accuracy but also on developing deployable models suitable for real building environments. The emergence of the hybrid metaheuristic ANN cluster indicates a transition from fixed-architecture prediction models toward adaptive optimization-based frameworks. Recent studies in other engineering domains have demonstrated the potential of nature-inspired optimization algorithms for feature selection and hyperparameter tuning. However, within chiller energy prediction, such approaches remain less mature compared with general energy forecasting applications, suggesting an opportunity for further investigation. Together, these clusters indicate a growing trend toward integrating evolutionary optimization with machine learning to improve building energy prediction accuracy. This evolution reflects a broader shift in the field from developing prediction models based primarily on algorithmic accuracy toward designing adaptive and application-oriented frameworks. The increasing integration of optimization techniques suggests that future chiller prediction models are expected to address multiple objectives, including prediction accuracy, computational efficiency, input variable selection, and operational applicability within building energy management systems.

It should be noted that the bibliographic coupling network is influenced by citation accumulation patterns. The minimum threshold of 68 citations was applied to identify publications with established citation relationships and to maintain the interpretability of the network. However, this criterion may reduce the visibility of recent publications,

particularly those published between 2023 and 2025, which may not have accumulated sufficient citations despite representing emerging research contributions. Therefore, the coupling clusters should be interpreted as representing the established intellectual structure of the field rather than a complete representation of the latest research activities. Recent studies were also considered during the analysis of research gaps and future directions in Section 4.4 to capture emerging trends beyond the citation-based network.

4.3. Thematic Evolution: Co-word Analysis

The co-word analysis identified four thematic clusters that complement and extend the intellectual structure identified through bibliographic coupling. The high total link strength of 5106 across 1257 links confirms strong conceptual interconnection across the four themes, indicating that research in this domain draws on a shared and stable vocabulary rather than fragmented or isolated topic areas.

The dominance of 'optimization' as the highest-frequency and most strongly connected keyword in the entire network (occurrences = 138, TLS = 646) is a notable finding. It indicates that the field is not purely concerned with passive energy consumption prediction but is oriented toward active optimization of building energy systems, where prediction models serve as components within broader decision-making and control frameworks. This is consistent with the applied engineering context of the domain, where building operators and facility managers require actionable outputs, not only accurate forecasts [70].

The co-word analysis revealed four major thematic clusters. Cluster 1 groups optimization, metaheuristic algorithms, HVAC systems, model predictive control, and thermal comfort, highlighting the widespread use of evolutionary computation techniques for energy management, control optimization, and multi-objective decision-making in buildings. Cluster 2 focuses on machine learning and cooling load prediction, including ANN, deep learning, random forest, and support vector regression, indicating that supervised learning remains the most active methodological area in building energy forecasting. The coexistence of multiple algorithms suggests that no single approach has emerged as the dominant solution, making comparative evaluations a common practice.

Cluster 3 occupies a central bridging role through keywords such as model, prediction, building energy, and consumption, linking optimization and machine learning research streams. This position underscores the importance of building energy prediction as a core application area that integrates computational intelligence with energy systems knowledge. Cluster 4 represents the interface between simulation-based and data-driven approaches, grouping keywords related to simulation, design, performance, and energy consumption. The growing use of simulation tools to generate training data for machine learning models suggests increasing integration between physics-based modeling and data-driven prediction, particularly when operational datasets are limited.

4.4. Research Gaps and Future Research Directions

Several research gaps emerge from the combined findings of the bibliographic coupling and co-word analyses. These gaps are discussed below in order of their relevance to the focus of the present study.

4.4.1. Feature selection for chiller-specific energy prediction

Although 'feature selection' does not appear among the top-ranked keywords in the co-word network, its methodological importance is well-established in the broader building energy prediction literature [71], [72]. The bibliographic coupling analysis shows that hybrid metaheuristic ANN approaches for feature selection and hyperparameter optimization (Cluster 4) represent an active and growing research front. Recent forecasting studies in other energy domains have further demonstrated the effectiveness of metaheuristic algorithms for feature selection and hyperparameter tuning, highlighting the potential of similar approaches for improving chiller-specific prediction models [73]. However, the application of advanced metaheuristic feature selection methods specifically to chiller systems, as opposed to whole-building or HVAC-level prediction, remains relatively underexplored. Most studies in the corpus address general building energy or HVAC-level prediction, with fewer studies targeting chiller-specific operational parameters such as return air humidity, supply air temperature, and saturation temperature as input features.

4.4.2. Interpretability and explainability of prediction models

The co-word analysis does not reveal any significant presence of terms related to explainable AI, model interpretability, or uncertainty quantification in the keyword network. This absence suggests that the majority of studies in this corpus have focused on maximizing predictive accuracy without systematically addressing the interpretability of model outputs for building operators and facility managers. Given the increasing expectation from building stakeholders that AI-based tools provide transparent and auditable recommendations, the development of interpretable prediction models for chiller energy management represents an important direction for future work. Post-hoc explanation methods including LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive Explanations) have demonstrated clear utility in adjacent building energy domains, most notably in the explanation of HVAC control policies, yet their systematic application to chiller energy prediction models remains conspicuously underexplored, suggesting that the chiller energy modelling literature has not fully capitalized on the explainability advances already established in the broader building energy management field [74].

4.4.3. Real-time deployment and computational efficiency

The co-word network does not contain keywords related to real-time deployment, edge computing, or computational latency, indicating that the translation of laboratory-validated prediction models to operational building management systems remains an open challenge. Most studies in the corpus evaluate model performance under offline batch-processing conditions using historical datasets, without addressing the computational requirements and data pipeline constraints associated with real-time prediction in operational settings [75]. Future research should consider the trade-off between model complexity and inference speed, particularly for chiller systems where control decisions may need to be made at 15-minute or shorter intervals.

4.4.4. Generalizability across building types and climatic regions

The geographical distribution of publications, with China accounting for nearly 40% of the corpus, raises questions about the generalizability of existing models to other climatic and operational contexts. The majority of chiller energy prediction studies have been conducted in temperate or continental climates, with relatively few studies addressing tropical or hot-arid climates where chiller systems operate under different load profiles and ambient conditions. As noted by [41], data-driven models trained on data from one building or climate zone may not transfer reliably to buildings in different contexts without retraining or transfer learning adaptation. Expanding the geographic and typological diversity of training datasets is therefore an important consideration for improving the practical applicability of chiller prediction models.

4.4.5. Integration of metaheuristic optimization with deep learning architectures

The co-word analysis shows that 'deep learning' and 'genetic algorithm' or 'multiobjective optimization' rarely appear in the same cluster, suggesting that the integration of metaheuristic optimization with deep learning architectures for building energy prediction remains relatively limited compared to the more common combination of metaheuristics with shallow ANN and MLP models. Given the demonstrated effectiveness of metaheuristic feature selection and hyperparameter tuning for

ANN-based building energy prediction [64], extending these approaches to deep learning architectures such as LSTM and transformer-based models represents a promising direction for improving prediction accuracy under complex, time-dependent operating conditions. Recent studies in other engineering domains have also demonstrated the integration of metaheuristic optimization with deep learning architectures to improve prediction performance, suggesting potential opportunities for similar approaches in chiller energy prediction applications [76]. The identified research gaps are interconnected and should be addressed through integrated research strategies rather than isolated methodological improvements. For example, feature selection influences computational efficiency, interpretability affects user acceptance of AI-based energy management tools, and climatic generalization determines whether prediction models developed in specific regions are transferable to other operating environments. Therefore, future advances in chiller energy prediction require consideration of both algorithmic development and practical deployment requirements.

4.5. Implications

The findings provide guidance for researchers and practitioners involved in building energy management, chiller optimization, and intelligent building operation. Beyond identifying publication patterns, the bibliometric analysis reveals how methodological developments in machine learning and optimization are connected with practical requirements in commercial building applications. The identified research clusters offer a clear overview of the field's development and emerging directions, helping researchers position future studies and focus on underexplored areas such as feature selection, model interpretability, and cross-building generalizability. The strong influence of review articles also highlights the need for further systematic reviews and meta-analyses that consolidate evidence across diverse prediction methods and applications.

From a practical standpoint, the results confirm that prediction and optimization are closely linked objectives in building energy research. Chiller prediction models should therefore be developed as components of broader optimization and control frameworks rather than standalone forecasting tools. The study also emphasizes the importance of feature selection for reducing model complexity and improving reliability in real-world systems. In addition, the limited contribution from Southeast Asian and African institutions suggests opportunities for targeted funding and research initiatives to address regional challenges and expand the global knowledge base on chiller energy prediction.

Despite the progress identified in this study, several practical challenges remain before machine learning-based chiller prediction models achieve widespread deployment in commercial buildings. Data privacy is an important consideration because building management systems collect detailed operational information, including occupancy patterns, equipment status, and energy consumption profiles. Appropriate data governance practices are required to ensure secure handling and responsible use of building data. In addition, prediction performance depends strongly on the quality and reliability of sensor measurements. Missing data, sensor drift, calibration errors, and communication failures may reduce model accuracy during long-term operation.

Computational cost and scalability are also important factors for practical deployment. Highly complex models, particularly deep learning architectures, may require substantial computational resources and extensive training data, which may limit implementation in buildings with restricted computing infrastructure. Future research should therefore investigate lightweight and adaptive models that maintain prediction accuracy while reducing computational requirements. Furthermore, trained models should be evaluated across different building types, climatic conditions, and operational scenarios to improve scalability and generalizability. Integration with existing building management systems should also be considered to ensure that prediction models provide actionable information for energy optimization and control strategies.

4.6. Limitations

First, the bibliometric dataset was retrieved exclusively from the Web of Science (WoS) Core Collection. Although WoS is widely recognized as a reliable source for bibliometric research due to its standardized metadata structure and established citation indexing practices [39], it does not represent the complete publication output of this research domain. Therefore, studies published in conference proceedings, technical reports, or journals indexed exclusively in Scopus, IEEE Xplore, or Google Scholar may not have been captured in the present analysis. The selection of a single database was motivated by the need to maintain data consistency, citation reliability, and compatibility with VOSviewer for reproducible network analysis. Nevertheless, this approach represents a trade-off between methodological consistency and broader literature coverage. Future bibliometric studies should consider integrating multiple databases with appropriate data harmonization and deduplication procedures to provide a more comprehensive representation of research trends in machine learning and metaheuristic optimization for chiller energy prediction. Second, the use of a keyword-based search string introduces selection bias. Some relevant documents using different terminology for chiller energy prediction or feature selection may have been excluded, while some tangentially related documents may have been included. The iterative refinement of search strings described in Section 2.2 mitigates this risk but does not eliminate it entirely.

Third, bibliographic coupling reflects the shared reference structure of documents at the time of data retrieval and may not fully capture the most recent intellectual developments in a rapidly evolving field. The citation threshold applied in VOSviewer (minimum 68 citations) limits the network to established works and necessarily excludes more recent high-quality publications that have not yet accumulated sufficient citations to meet the threshold, including some papers published in 2023, 2024, and 2025.

Fourth, the labeling of cluster themes involves a degree of interpretive judgment based on reviewing the titles, abstracts, and keywords of representative documents within each cluster. This is a recognized limitation of bibliometric analysis and does not affect the validity or reproducibility of the quantitative network analysis itself [33]. Researchers with different domain knowledge may assign somewhat different thematic labels to the same clusters, though the underlying network structure remains objective and replicable. The principal contribution of this study extends beyond descriptive bibliometric statistics. By integrating bibliographic coupling with keyword co-occurrence analysis, this work reveals how research has evolved from conventional ANN-based prediction toward hybrid optimization-driven learning frameworks while identifying unresolved challenges in feature selection, interpretability, deployment, and climatic generalization. This focused mapping provides a reference framework for researchers seeking future research directions in machine learning-based chiller energy prediction.

5. Conclusion

This study presents a bibliometric analysis of 419 Web of Science journal articles published between 2016 and 2026 on machine learning, metaheuristic optimization, and chiller energy prediction. Using bibliographic coupling and keyword co-occurrence analysis in VOSviewer, the study examined the intellectual structure, thematic evolution, and emerging research directions of the field. Results show a strong growth trend, increasing from 9 publications in 2016–75 in 2025, with 20,097 citations and an h-index of 63. China emerged as the leading contributor, while 66 countries participated in this research area. Four major research fronts were identified, covering ANN-based predictive control and HVAC optimization, deep learning and transfer learning for

energy forecasting, foundational review and benchmarking studies, and hybrid machine learning-metaheuristic approaches for heating and cooling load prediction. Keyword analysis further revealed dominant themes related to optimization, intelligent control, machine learning, building energy prediction, and simulation-based performance assessment.

The analysis identified several important research gaps. Chiller-specific feature selection using advanced metaheuristic algorithms remains limited, while model interpretability, explainable AI, real-time deployment, and edge computing receive little attention. The concentration of studies in specific geographic regions also raises concerns regarding model generalizability across diverse climatic conditions. In addition, the integration of deep learning with metaheuristic optimization for chiller prediction remains underdeveloped compared with traditional ANN-based approaches. These findings provide a structured overview of the field and suggest future research directions, including the development of interpretable hybrid models, real-time implementation in building management systems, and comprehensive benchmarking of feature selection and optimization techniques across different chiller configurations and climates.

CRediT authorship contribution statement

Nor Farizan Zakaria: Resources, Project administration, Methodology, Visualization, Formal analysis, Writing – original draft. **Mohd Redzuan Ahmad:** Investigation, Data curation. **Zuriani Mustaffa:** Validation, Writing – review & editing. **Mohd Herwan Sulaiman:** Supervision, Conceptualization, Writing – Reviewing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data generated and used for this study was obtained from primary sources and was duly analyzed by the authors. The data is not bound to any copyright of any organization or establishment. Data can be shared upon request at any time decided by the editor(s).

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